Reporting No: 03 Week No:03 From : 4/08/26 To:10/08/26
Project ID: 34
Project Title: SignBridge-AI

WEEKLY REPORT 3

Work done in last week (Attach supporting Documents):

Sign Recognition Model Development, Training & Evaluation

1. Deep Learning Model Research

- Studied deep-learning approaches suitable for sign-language sequence recognition.
- Reviewed different sequence-model architectures, including:
  - RNN
  - LSTM
  - GRU
  - Bidirectional LSTM
  - Transformer-based sequence models
- Studied why temporal models are required for sign-language recognition.
- Understood the difference between:
  - Image classification
  - Video classification
  - Sequence classification
- Selected LSTM as the initial baseline model because sign-language data consists of sequences of landmark frames and LSTM can learn temporal relationships between consecutive frames.

2. Dataset Preparation for Model Training

- Used the landmark sequences prepared during Week 2 as input to the deep-learning pipeline.
- Converted processed landmark data into model-compatible numerical arrays.
- Prepared class labels corresponding to individual signs.
- Verified:
  - Input dimensions
  - Sequence length
  - Number of features
  - Number of classes
- Organized the dataset into:
  - Training set
  - Validation set
  - Testing set
- Ensured that the temporal order of landmark frames was preserved.

The model input pipeline was established as:

Sign Video
↓
Frames
↓

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MediaPipe Landmarks
↓
Preprocessing
↓
Landmark Sequence
↓
Training Dataset
↓
LSTM Model

3. LSTM Model Implementation

- Implemented the initial LSTM-based sign-recognition architecture using PyTorch.
- Designed the model to accept sequential landmark features.
- Implemented:
  - Input layer
  - LSTM layer(s)
  - Hidden representation
  - Fully connected classification layer
  - Output layer
- Configured the model according to the number of sign classes in the prepared dataset.
- Studied the role of:
  - Hidden states
  - Time steps
  - Batch size
  - Sequence length
  - Output classes

The initial architecture was designed conceptually as:

Landmark Sequence
↓
Input Layer
↓
LSTM
↓
Hidden Representation
↓
Fully Connected Layer
↓
Softmax / Class Scores
↓
Predicted Sign

4. Model Training Pipeline

- Developed the initial training pipeline.
- Configured the training components:
  - Loss function

- Optimizer
- Learning rate
- Batch size
- Number of epochs

- Implemented training and validation loops.
- Recorded training metrics after each epoch.
- Monitored:
  - Training loss
  - Validation loss
  - Training accuracy
  - Validation accuracy
- Saved the trained model for later inference.

The training workflow was:

Training Dataset
↓
DataLoader
↓
LSTM
↓
Prediction
↓
Loss Calculation
↓
Backpropagation
↓
Optimizer Update
↓
Next Epoch

5. Landmark Extraction & Dataset Preparation

- Implemented landmark extraction from live webcam frames.
- Converted landmark data into numerical coordinates.
- Stored extracted landmarks in CSV format for future model training.
- Designed the initial dataset collection pipeline.
- Prepared documentation and project notes for future development.

Reason for incomplete work:

1.
Model training has not started because the first week focused on setting up the development environment and data collection pipeline.

2.
Large-scale public datasets (INCLUDE, WLASL, How2Sign) have not yet been integrated and will be addressed in Week 2.

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3.
Backend API, frontend interface, and AI models are scheduled for future development phases.

Plans for next week:

1. **Dataset Collection & Preprocessing**
   - Download and study public sign language datasets.
   - Convert videos into MediaPipe landmark sequences.
   - Clean and preprocess datasets for training.
2. **Feature Engineering**
   - Normalize landmark coordinates.
   - Generate training, validation, and testing datasets.
   - Perform data augmentation.
3. **AI Model Preparation**
   - Design the LSTM-based sign recognition model.
   - Prepare the training pipeline using PyTorch.
   - Evaluate preprocessing techniques for improved recognition accuracy.

References:

- Python Official Documentation
- OpenCV Documentation
- MediaPipe Official Documentation
- PyTorch Documentation
- Git & GitHub Documentation
- NumPy Documentation
- Pandas Documentation
- WLASL Dataset
- INCLUDE Indian Sign Language Dataset
- How2Sign Dataset

Signature of External Guide **Signature of Internal Guide**

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